

# Mathematics for Computer Science

COMP1021: Durham University

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# Part I

## Calculus

### 1 Introduction to Calculus

#### 1.1 Sets

**Definition 1.1.1:** A set is an unordered collection of distinct elements denoted as  $A = \{x | \text{condition}\}$ .

Some common set operations include:

- **Definition 1.1.2:** The union of two sets,  $A \cup B$ , includes all the elements of both sets.
- **Definition 1.1.3:** The intersection of two sets,  $A \cap B$ , includes any elements which are present in both sets.
- **Definition 1.1.4:** If  $A$  and  $B$  are finite sets, then the principle of inclusion-exclusion states that

$$A \cup B = A + B - (A \cap B) \quad (1)$$

- **Definition 1.1.5:** The set  $A$  is a subset of  $B$ ,  $A \subseteq B$ , if every element of  $A$  is an element of  $B$ .  $A$  is a proper subset of  $B$ ,  $A \subset B$ , if  $A$  is a subset of  $B$  and there is at least one element of  $B$  not in  $A$ .
- **Definition 1.1.6:** The powerset of a set  $A$  is the set of all subsets of  $A$ .

$$\mathcal{P}(A) = \{A' | A' \subseteq A\} \quad (2)$$

- **Definition 1.1.7:** The cartesian product of two sets,  $A \times B$ , is the set of all possible ordered pairs with the first element from  $A$  and the second from  $B$ .

**Definition 1.1.8:** For a finite set, the cardinality is the number of distinct elements that it contains.

#### 1.2 Functions

For a function  $f : A \rightarrow B$ ,

- **Definition 1.2.1:**  $A$  is known as the domain of  $f$ .
- **Definition 1.2.2:**  $B$  is known as the codomain of  $f$ .
- **Definition 1.2.3:** If  $f(a) = b$ , then  $b$  is the image of  $a$ .
- **Definition 1.2.4:** The pre-image of  $b$  is the set  $\{a | f(a) = b\}$ .
- **Definition 1.2.5:**  $f$  forms an injection if every element of  $A$  maps to a unique element in  $B$ , even if some elements of  $B$  don't have a mapping.
- **Definition 1.2.6:**  $f$  forms a surjection if every element of  $B$  maps to at least one element of  $A$ .
- **Definition 1.2.7:**  $f$  forms a bijection if it is both injective and surjective.

**Definition 1.2.8:** If for an unbounded set  $A$ , there is a bijective function  $f : A \rightarrow \mathbb{N}$ , then we say  $A$  is countably infinite, otherwise  $A$  is known as uncountably infinite.

### 1.3 Hyperbolic Functions

**Definition 1.3.1:** Given the unit hyperbola  $x^2 - y^2 = 1$ , the hyperbolic functions are defined by the coordinates of a point at which area  $\frac{1}{2}\alpha$  has been swept out.

$$\begin{aligned} \sinh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{2} & \cosh(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{2} & \tanh(\alpha) &= \frac{e^\alpha - e^{-\alpha}}{e^\alpha + e^{-\alpha}} & \sigma(\alpha) &= \frac{1}{1 + e^{-\alpha}} \\ \operatorname{sech}(\alpha) &= \frac{2}{e^\alpha + e^{-\alpha}} & \operatorname{cosech}(\alpha) &= \frac{2}{e^\alpha - e^{-\alpha}} & \coth(\alpha) &= \frac{e^\alpha + e^{-\alpha}}{e^\alpha - e^{-\alpha}} \end{aligned}$$

These functions can then be differentiated as follows:

$$\begin{aligned} \frac{d}{dx} [\sinh(\alpha)] &= \cosh(\alpha) & \frac{d}{dx} [\cosh(\alpha)] &= \sinh(\alpha) \\ \frac{d}{dx} [\tanh(\alpha)] &= \operatorname{sech}^2(\alpha) & \frac{d}{dx} [\sigma(\alpha)] &= \sigma(\alpha)(1 - \sigma(\alpha)) \end{aligned}$$

## 2 Multivariate Functions

### 2.1 Multivariate Derivatives

**Definition 2.1.1:** For a function  $f(x, y)$ , if we take  $f = f_0$  to be fixed but unknown, then we can differentiate  $f$  as a function of  $x$  at  $x = x_0$ , giving us the partial derivative of  $f$  with respect to  $x$  which gives the gradient when moving in the  $x$ -direction from any point.

$$\frac{\delta f}{\delta x} [f(x, y)] = \lim_{x \rightarrow x_0} \left( \frac{f(x, y_0) - f(x_0, y_0)}{x - x_0} \right) = f_x \quad (3)$$

**Definition 2.1.2:** The vector of partial derivatives, denoted as  $\nabla f$ , contains all the partial derivatives of a function. This forms a vector pointing in the direction of the greatest rate of increase of  $f$ , where its magnitude is the rate of increase in that direction. For example, for a function  $f(x, y)$ ,

$$\nabla f = \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad (4)$$

**Definition 2.1.3:** The directional derivative of the function  $f(x, y)$  gives the gradient at any point whilst looking in the direction given by the unit vector  $\mathbf{v} = \begin{pmatrix} v_x \\ v_y \end{pmatrix}$ .

$$\nabla_{\mathbf{v}} f = \lim_{h \rightarrow 0} \frac{f(x_0 + h\mathbf{v}) - f(x_0)}{h} \quad (5)$$

If the slope is approximately flat at a small scale, then the slope can be determined by

$$\nabla_{\mathbf{v}} f = \mathbf{v} \cdot \nabla f \quad (6)$$

From this, we get that, for a function  $f(\mathbf{x}_0)$  and a direction  $\mathbf{v}$ ,

$$f(\mathbf{x}_0 + h\mathbf{v}) \approx f(\mathbf{x}_0) + h\mathbf{v} \cdot \nabla f(\mathbf{x}_0) \quad (7)$$

## 2.2 Multivariate Extrema

**Definition 2.2.1:** A function that is continuous is in class  $C^0$ , and a function that is continuous and has a continuous derivative is in class  $C^1$ . A function  $f$  such that  $f'$ ,  $f''$ , ...,  $f^{(k)}$  exist and are continuous is in class  $C^k$ .

**Definition 2.2.2:** A function  $f$  such that derivatives of all orders exist and are continuous is in class  $C^\infty$  is also smooth.

**Definition 2.2.3:** All the second-order partial derivatives of a function  $f(x, y)$  can be gathered into a Hessian matrix.

$$\mathbf{H}_f = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} \quad (8)$$

This can be used to determine the form of bivariate extrema. Suppose  $f(x, y) \in C^2$  and  $f_x(x_0, y_0) = 0 = f_y(x_0, y_0)$  then,

- $(x_0, y_0)$  is a local maximum if  $\det(\mathbf{H}_f) > 0$  and  $f_{xx} < 0$  at  $(x_0, y_0)$ .
- $(x_0, y_0)$  is a local minimum if  $\det(\mathbf{H}_f) > 0$  and  $f_{xx} > 0$  at  $(x_0, y_0)$ .
- $(x_0, y_0)$  is a saddle point if  $\det(\mathbf{H}_f) < 0$ .
- If  $\det(\mathbf{H}_f) = 0$ , then the test is inconclusive.

At a stationary point  $x_0$ , the quadratic approximation of a function  $f(x)$  is

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} \mathbf{v}^T \mathbf{H}_f(x_0) \mathbf{v} \quad (9)$$

But as  $\mathbf{H}_f(x_0)$  is real and symmetric, it has orthogonal eigenvectors  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$  with eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$ , so we can write

$$\mathbf{v} = c_1 \mathbf{e}_1 + c_2 \mathbf{e}_2 + \dots + c_n \mathbf{e}_n \quad (10)$$

When substituted back into the original equation, this gives that

$$f(x_0 + \mathbf{v}) \approx f(x_0) + \frac{1}{2} (\lambda_1(c_1)^2 + \lambda_2(c_2)^2 + \dots + \lambda_n(c_n)^2) \quad (11)$$

Given this, we can use the eigenvalues to determine the nature of the stationary point  $x_0$ ,

- If all eigenvalues are positive, then the stationary point is a local minimum.
- If all eigenvalues are negative, then the stationary point is a local maximum.
- If there is a mixture of positive and negative eigenvalues, then the stationary point is a saddle point.

## 2.3 Vector-Valued Functions

**Definition 2.3.1:** A vector-valued function takes an  $n$ -dimensional vector as input, and outputs an  $m$ -dimensional vector, e.g.,  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ .

**Definition 2.3.2:** The Jacobian matrix of a vector-valued function  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$  can be formed by setting  $\mathbf{J}_{i,j} = \frac{\delta f_i}{\delta x_j}$  to form,

$$\mathbf{J}_f = \begin{pmatrix} \frac{\delta f_1}{\delta x_1} & \dots & \frac{\delta f_1}{\delta x_n} \\ \vdots & \ddots & \vdots \\ \frac{\delta f_m}{\delta x_1} & \dots & \frac{\delta f_m}{\delta x_n} \end{pmatrix} \quad (12)$$

For a multivariate function  $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ , the Jacobian matrix can be used to linearly approximate  $f$  at  $x$ ,

$$f(x + h\mathbf{v}) \approx f(x) + \mathbf{J}_f(x)(h\mathbf{v}) \quad (13)$$

## 3 Lagrange Multipliers

### 3.1 Constrained Optimisation

**Definition 3.1.1:** The goal of a constrained optimisation problem is to either minimise or maximise a multivariate function  $f(\mathbf{x})$  where  $x_1, x_2, \dots, x_n$  are not independent subject to some constraint  $g(\mathbf{x})$ .

**Definition 3.1.2:** The Lagrange multipliers method is used to find stationary points of an objective function  $f(\mathbf{x}) \in C^1$  subject to the constraint  $g(\mathbf{x}) = 0$ .

1. Define a new objective function called the Lagrangian,

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda g(\mathbf{x}) \quad (14)$$

2. Find the stationary points of  $\mathcal{L}$  with respect to both  $\mathbf{x}$  and  $\lambda$ , i.e., solve the system of equations  $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$  and  $g(\mathbf{x}) = 0$  for the unknowns  $\mathbf{x}$  and  $\lambda$ .
3. Then, the constrained extrema are found among the solutions to these equations. Identify their nature.

## 4 Series

### 4.1 Introduction to Series

**Definition 4.1.1:** A sequence is an ordered, countably infinite, set of numbers.

**Definition 4.1.2:** A series is the sum of the members of a sequence.

**Theorem 4.1.1:** If the series  $\sum_{n=0}^{\infty} a_n$  and the series  $\sum_{n=0}^{\infty} b_n$  are convergent, and have respective sums  $\alpha$  and  $\beta$ , then

$$\sum_{n=0}^{\infty} (a_n + b_n) = \alpha + \beta \quad (15)$$

and

$$\sum_{n=0}^{\infty} (a_n - b_n) = \alpha - \beta \quad (16)$$

*Proof:*  $\sum_{n=0}^{\infty} a_n$  converges to  $\alpha$  and  $\sum_{n=0}^{\infty} b_n$  converges to  $\beta$ . Given an arbitrary  $\epsilon > 0$ , let  $\epsilon_a = \frac{1}{2}\epsilon = \epsilon_b$ . Since  $\sum_{n=0}^{\infty} a_n$  converges,  $\exists N_a$  such that,

$$\left| \sum_{r=0}^n a_r - \alpha \right| < \epsilon_a, \quad \forall n > N_a \quad (17)$$

Similarly, since  $\sum_{n=0}^{\infty} b_n$  converges,  $\exists N_b$  such that,

$$\left| \sum_{r=0}^n b_r - \beta \right| < \epsilon_b, \quad \forall n > N_b \quad (18)$$

Taking  $N = \max\{N_a, N_b\}$ , then  $\forall n > N$ ,

$$\begin{aligned} \left| \sum_{r=0}^n (a_r + b_r) - (\alpha + \beta) \right| &= \left| \left( \sum_{r=0}^n a_r - \alpha \right) + \left( \sum_{r=0}^n b_r - \beta \right) \right| \\ &\leq \left| \sum_{r=0}^n a_r - \alpha \right| + \left| \sum_{r=0}^n b_r - \beta \right| \leq \epsilon_a + \epsilon_b = \epsilon \end{aligned} \quad (19)$$

i.e,  $\sum_{n=0}^{\infty} (a_n + b_n)$  converges to  $(\alpha + \beta)$  using triangle inequality  $|a + b| \leq |a| + |b|$ .

□

**Theorem 4.1.2:** If  $\lim_{n \rightarrow \infty} a_n \neq 0$  then the series  $\sum_{n=0}^{\infty} a_n$  diverges.

*Proof:* Suppose  $\sum_{n=0}^{\infty} a_n$  converges for some  $S$ , Then,  $\lim_{n \rightarrow \infty} S_n = S$ , which means that given arbitrary  $\epsilon > 0$ , there exists some  $N_\epsilon$  such that, for all  $n > N_\epsilon$ ,

$$|S_n - S| < \epsilon \quad |S_n - S| < 2\epsilon \quad (20)$$

Combining these,

$$2\epsilon > |S_n - S| + |S_{n+1} - S| = |S - S_n| + |S_{n+1} - S| \geq |S_{n+1} - S_n| = |a_{n+1}| \quad (21)$$

But  $\epsilon$  was arbitrary, so we have shown that, for any  $\epsilon' > 0$  there exists an  $N_\epsilon$  such that  $|a_n| < \epsilon'$  for all  $n > N_\epsilon$ .

□

As the number of elements in a series approaches infinity...

- **Definition 4.1.3:** If  $\lim_{n \rightarrow \infty} a_n \neq 0$ , then the series diverges.
- **Definition 4.1.4:** If  $\lim_{n \rightarrow \infty} a_n = 0$ , then the series converges.
- **Definition 4.1.5:** If  $\lim_{n \rightarrow \infty} |a_n| = 0$ , then the series absolutely converges.
- **Definition 4.1.6:** If  $\lim_{n \rightarrow \infty} a_n = S$ , then the series is convergent to  $S$ .

**Theorem 4.1.3:** The ratio test states that, if  $\sum_{n=1}^{\infty} a_n$  is a series where  $a_n \neq 0$  and  $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$  then

- If  $L < 1$ , the series absolutely converges.
- If  $L > 1$ , the series is divergent.
- If  $L = 1$ , then the test fails.

*Proof:* (a) If  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L < 1$ , then  $\exists r$  with  $L < r < 1$  and  $N$  such that  $\forall n > N$ :

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < r \quad \rightarrow \quad |a_{n+1}| < r|a_n| \quad (22)$$

Then  $|a_{n+2}| < r|a_{n+1}| < r^2|a_n|$  and  $|a_{n+3}| < r|a_{n+2}| < r^3|a_n|$  etc. So the remainder is

$$R_{n+m} = \lim_{m \rightarrow \infty} \left( |a_n| r^{m+1} \frac{1}{1-r} \right) = 0 \quad (23)$$

Hence the series converges.

□

**Definition 4.1.7:** The  $n^{\text{th}}$  root test states that, for the series  $\sum_{n=1}^{\infty} a_n$ ,  $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$ ,

- If  $L < 1$ , then the series absolutely converges.
- If  $L > 1$ , then the series is divergent.
- If  $L = 1$ , then the test fails.

**Definition 4.1.8:** The alternating series test states that the series  $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$  converges if  $a_n > 0$ ,  $a_{n+1} \leq a_n \forall n$ , and  $\lim_{n \rightarrow \infty} a_n = 0$

**Theorem 4.1.4:** Every subsequence of a convergent sequence converges to the same limit as the original sequence.

*Proof:* Let  $\{a_{n_k}\}$  be a subsequence of  $\{a_n\}$ . First note that  $n_k \geq k$  and so  $n_{k+1} \geq k+1$ . Let  $\lim_{n \rightarrow \infty} a_n = L$ . Then, for any  $\epsilon > 0$ , there exists  $N_\epsilon$  such that for all  $n > N_\epsilon$  it holds that  $|a_n - L| < \epsilon$ . Let  $k > N_\epsilon$ , then  $n_k > N_\epsilon$  and so it must be that  $|a_{n_k} - L| < \epsilon$ .

□

## 4.2 Power Series

**Definition 4.2.1:** A power series is a series involving a variable  $x$  and a constant  $x_0$  of the form

$$\sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \dots \quad (24)$$

At any specific value of  $x$ , the power series becomes a normal series.

**Definition 4.2.2:** For an arbitrary power series  $\sum_{n=0}^{\infty} a_n (x - x_0)^n$ , we can use the ratio test (including the  $(x - x_0)$  term). From this, we call  $\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = r$  the radius of convergence. If  $|x - x_0| < r$ , we have absolute convergence and if  $|x - x_0| > r$ , we have divergence.

Let  $f(x) = \sum_{n=0}^{\infty} a_n x^n$  be a power series with radius of convergence  $r > 0$ . Then, within the interval  $(-r, r)$ :

- $f(x)$  is continuous.
- $f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$  (term-by-term differentiation).
- $\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$  (term-by-term integration).
- Both series  $f'(x)$  and  $\int_0^x f(t) dt$  have the same radius of convergence as  $f(x)$ .

**Definition 4.2.3:** Suppose we start with some function  $f(x)$  and we wish to determine a power series  $\sum_{n=0}^{\infty} a_n x^n$  such that  $f(x) = \sum_{n=0}^{\infty} a_n x^n$ . Putting these values in, we see that if such a power series exists, then:

$$f(x) = f(0) + f'(0)x + f''(0)\frac{x^2}{2!} + f^{(3)}(0)\frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} f^{(n)}(0)\frac{x^n}{n!} \quad (25)$$

This is called the Maclaurin series for  $f$ . To use the series, we must have that  $f \in C^\infty$  and the power series converges.

**Definition 4.2.4:** We can recenter the power series on  $x_0$  by defining a new function  $g(x) = f(x + x_0)$  then  $g(0) = f(x_0)$  and  $g'(x) = f'(x + x_0)$ . Repeatedly differentiating,  $g^{(n)}(x) = f^{(n)}(x + x_0)$  so we build the Maclaurin series for  $g$ , which gives us

$$f(x + x_0) = g(x) = \sum_{n=0}^{\infty} g^{(n)}(0)\frac{x^n}{n!} = \sum_{n=0}^{\infty} f^{(n)}(x_0)\frac{x^n}{n!} \quad (26)$$

or,

$$f(x) = g(x - x_0) = f(x_0) + f'(x_0)(x - x_0) + f''(x_0)\frac{(x - x_0)^2}{2!} + \dots \quad (27)$$

This form is called the Taylor series expansion of  $f$ .

### 4.3 Taylor's Theorem

**Lemma 4.3.1:** If a real-valued function  $f$  is continuous on  $[a, b]$ , differentiable on  $(a, b)$ , and  $f(a) = f(b)$ , then there exists  $\xi \in (a, b)$  such that  $f'(\xi) = 0$ .

*Proof:* Not proved here. □

**Theorem 4.3.1:** Suppose a function  $f$  is  $n$  times differentiable on  $[a, x]$ , then there exists some  $\xi \in (a, x)$  such that

$$f(x) = f(a) + (x - a)f'(a) + \dots + \frac{(x - a)^{(n-1)}}{(n - 1)!}f^{(n-1)}(a) + R^n(x) \quad (28)$$

Where the remainder term is equal to  $R_n(x) = \frac{(x-a)^n}{n!}f^{(n)}(\xi)$ .

*Proof:* Suppose a function  $f$  is  $n$  times differentiable on  $[a, x]$ . Let  $k$  be a constant defined so that

$$f(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2!}f''(a) + \dots + \frac{(x - a)^{(n-1)}}{(n - 1)!}f^{(n-1)}(a) + \frac{(x - a)^n}{n!}k \quad (29)$$

We can then define the function

$$F(y) = f(x) - f(y) - (x - y)f'(y) - \dots - \frac{(x - y)^{(n-1)}}{(n - 1)!}f^{(n-1)}(y) - \frac{(x - y)^n}{n!}k \quad (30)$$

Then  $F(x) = 0$  and also  $F(a) = f(x) - f(x) = 0$  so we can apply Rolle's theorem to  $F$  on  $[a, x]$ , meaning there is some  $\xi \in (a, x)$  such that  $F'(\xi) = 0$ . □

### 4.4 Fourier Series

## Part II

# Linear Algebra

## Part III

# References

### 5 Module Professors

The below is accurate as of the 2025/26 academic year and will not be updated in subsequent years.

- *Calculus* is taught by Doctor Junyan Hu and Doctor Sukanya Pandey.

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- *Linear Algebra* is taught jointly by Professor Steven Bradley and Doctor Karl Southern.

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### 6 Cited Works